# Constraint Programming - MiniZinc Practice
## The Estates

Declarative Programming, Applications.
December 16, 2025

## Introduction

A construction company has specialized in high-end homes with a fixed floor plan. Essentially, what it does is take 10×10 meter squares and turn them into plots with 5 or 6 of these squares. In Figure 1, you will find the shapes of the 500 $m2$. For the 600 $m2$ , only the 2×3 rectangular shape is currently offered. Note that some of these plots can be rotated, and some can even be flipped.

The combinatorial problem arises when the company encounters the whims of millionaires who ask for these original shapes and has to place them on the specific plot of land they have to build on, all while respecting certain style restrictions: for example, a particular neighbor doesn't want to see them, or a certain part of the lot is occupied by parks or paths.

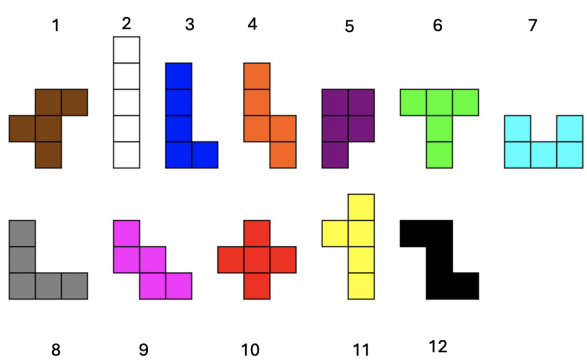


Figure 1: 5-square plot shapes.

In Figure 2, we find a solution that, on a plot of 7×7 squares of 100 $m2$, and where construction was not allowed in the areas reserved for parks and paths, has placed 6 plots of 500 $m2$ of types 3, 4, 5, 7, 8, 11, and 12 and one of 600 $m2$ (let's call it type 13). In this case, it was known that the first plot (the type 3 one) could not touch the last plot (the type 13 one), and the fourth plot (the type 7 one) had to have a park on at least one side.

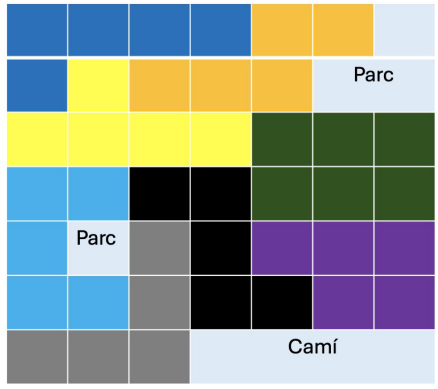


Figure 2: Estates laid out in $7x7$ plots, with 8 squares reserved for roads and parks. Also satisfying neighborhood restrictions and the park privilege for some estate.

## It is requested

You are asked to create a model in MiniZinc and design a data format to specify problems of this type. Specifically, your model must be able to solve a problem like the one in the example in a reasonable amount of time, ideally in seconds.

## Data Model

Your data model (.dzn) will need to specify:

- Dimensions of the buildable lot (in the example, it was $7 \times 7$).
- Parcels designated for parks.
- Parcels designated for roads.
- Parcels to be placed, indicating their type (e.g., a list of numbers from 1 to 13, where 13 would refer to the $2 \times 3$ parcel).
- Pairs of parcels that cannot be adjacent (pairs of indices from the list above).
- Squares that must have at least one park square as a neighbor.

## MiniZinc Model

Your MiniZinc (.mzn) model must be able to solve the instances you pass it in a reasonable amount of time and display the solution on screen in a readable and understandable way, ideally by providing a "representation" of what the parcel would look like when built.

Figure 3: Estates divided into $3x7$ plots.

## Experimentation

Start with a simple example like the one in Figure 3 and make it more complicated. $7x7$ instances with 8 squares reserved for paths and parks but without adjacency restrictions or park privileges should be easily solvable. Can we move on to $8 \times 8$ instances? How far can you get?